

PROJECT REPORT

HEART DISEASE PREDICTION USING MACHINE LEARNING

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ABSTRACT

Heart disease is one of the leading causes of death worldwide and poses a significant challenge to healthcare systems. Early diagnosis and timely treatment play a crucial role in reducing mortality and improving patient outcomes. With the rapid growth of healthcare data and advancements in artificial intelligence, machine learning has emerged as an effective tool for predicting diseases based on historical medical records.

This project focuses on developing a Heart Disease Prediction System using machine learning techniques. The objective is to analyze patient health records and build a classification model capable of predicting whether a patient is likely to suffer from heart disease. The dataset contains several clinical and demographic attributes, including age, gender, chest pain type, resting blood pressure, cholesterol level, fasting blood sugar, resting electrocardiographic results, maximum heart rate achieved, exercise-induced angina, ST depression, slope of the ST segment, number of major vessels, thalassemia, and the target variable representing the presence or absence of heart disease.

The project begins with loading the dataset into Python using the Pandas library, followed by data preprocessing to handle missing values, identify duplicate records, and prepare the data for analysis. Exploratory Data Analysis (EDA) is performed using statistical methods and graphical visualizations to understand feature distributions and identify relationships among variables.

Multiple machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM) are considered for classification. Their performance is evaluated using standard evaluation metrics including Accuracy, Precision, Recall, F1-score, Confusion Matrix, and ROC-AUC score. Feature importance analysis is also performed to identify the medical factors that contribute most significantly to heart disease prediction.

The results demonstrate that machine learning techniques can effectively predict heart disease with high accuracy while providing valuable insights into the factors influencing cardiovascular health. Although these predictive models cannot replace professional medical diagnosis, they can serve as decision-support tools that assist healthcare professionals in identifying high-risk patients. Future improvements may include larger datasets, hyperparameter optimization, deep learning models, and deployment as a web-based healthcare application.

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1. INTRODUCTION

Heart disease is one of the most serious health problems worldwide and is responsible for millions of deaths every year. It includes several cardiovascular conditions such as coronary artery disease, heart attack, heart failure, and arrhythmia. Early diagnosis of heart disease is extremely important because timely medical intervention can significantly reduce complications and improve patient survival rates.

With the rapid growth of healthcare data, hospitals and medical institutions generate enormous amounts of patient information every day. Analyzing this data manually is both time-consuming and prone to human error. Machine Learning (ML), a branch of Artificial Intelligence (AI), provides efficient techniques for analyzing medical data and identifying patterns that may indicate the presence of heart disease.

Machine learning algorithms can learn from historical patient records and build predictive models capable of classifying whether a patient is likely to have heart disease. Such prediction systems assist doctors by providing quick and accurate assessments, thereby improving clinical decision-making. However, these systems are intended to support healthcare professionals rather than replace medical diagnosis.

This project focuses on analyzing a Heart Disease dataset using Python and its data science libraries. The dataset contains various medical attributes including age, sex, chest pain type, resting blood pressure, cholesterol level, fasting blood sugar, electrocardiographic results, maximum heart rate achieved, exercise-induced angina, ST depression, slope of the ST segment, number of major vessels, and thalassemia. These features are used to develop machine learning models capable of predicting heart disease.

The project includes data preprocessing, exploratory data analysis, visualization, model development, and performance evaluation. The final outcome demonstrates how machine learning techniques can contribute to early disease detection and improve healthcare services.

2. OBJECTIVES

The primary objective of this project is to develop a machine learning model capable of predicting heart disease using patient medical data. The project also aims to understand the relationships among different clinical attributes through statistical analysis and visualization.

The specific objectives of this project are:

- To load the Heart Disease dataset and understand its structure.
- To identify the number of records, features, and target classes.
- To perform exploratory data analysis (EDA).
- To identify and handle missing values and duplicate records.
- To analyze the statistical properties of each feature.
- To visualize data distributions using histograms and box plots.
- To study relationships among variables using correlation analysis.
- To split the dataset into training and testing datasets.
- To build machine learning classification models.
- To compare the performance of different algorithms.
- To evaluate models using Accuracy, Precision, Recall, F1-score, ROC-AUC Score, and Confusion Matrix.
- To identify the most influential features affecting heart disease prediction.
- To summarize the findings and discuss future improvements.

By achieving these objectives, the project demonstrates the usefulness of Artificial Intelligence in assisting medical professionals with early diagnosis and healthcare decision-making.

3. METHODOLOGY

3.1 TOOLS USED

The project is implemented using Python and various data science libraries that simplify data preprocessing, visualization, and machine learning.

Python

Python is the primary programming language used for developing the project due to its simplicity and extensive machine learning ecosystem.

Jupyter Notebook

Jupyter Notebook provides an interactive environment for writing Python code, executing data analysis, and visualizing outputs.

Pandas

Pandas is used for loading, cleaning, manipulating, and analyzing the Heart Disease dataset.

NumPy

NumPy provides numerical computing functions and efficient array operations required during preprocessing.

Matplotlib

Matplotlib is used for creating graphs such as histograms, scatter plots, bar charts, and line charts.

Seaborn

Seaborn provides advanced statistical visualizations including heatmaps, box plots, and pair plots.

Scikit-learn

Scikit-learn provides machine learning algorithms, preprocessing utilities, model evaluation metrics, and train-test splitting functions.

3.2 DATASET DESCRIPTION

The Heart Disease dataset contains patient medical information collected from clinical examinations. Each row represents one patient, while each column represents a medical feature that may influence heart disease diagnosis.

The dataset includes the following attributes:

Feature	Description
Age	Age of the patient
Sex	Gender of the patient
Chest Pain Type (cp)	Type of chest pain experienced
Resting Blood Pressure (trestbps)	Blood pressure at rest
Cholesterol (chol)	Serum cholesterol level
Fasting Blood Sugar (fbs)	Blood sugar level after fasting
Resting ECG (restecg)	Electrocardiographic results
Maximum Heart Rate (thalach)	Highest heart rate achieved
Exercise Induced Angina (exang)	Exercise-related chest pain
Oldpeak	ST depression induced by exercise
Slope	Slope of the peak exercise ST segment
CA	Number of major vessels
Thal	Thalassemia condition
Target	Presence (1) or Absence (0) of Heart Disease

The target variable represents whether the patient has heart disease and is used for classification.

3.3 DATA LOADING

The dataset is imported into Python using the Pandas library.

After loading the dataset, the following operations are performed:

- Displaying the first few records.
- Checking the dataset dimensions.
- Displaying column names.
- Identifying data types.
- Checking missing values.
- Detecting duplicate records.
- Displaying summary statistics.

These preliminary analyses help ensure that the dataset is complete and suitable for machine learning.

3.4 DATA PREPROCESSING

Data preprocessing is an essential stage because machine learning models require clean and consistent data.

The preprocessing steps include:

- Handling missing values if present.
- Removing duplicate observations.
- Encoding categorical variables where necessary.
- Splitting the dataset into input features (X) and target variable (Y).
- Dividing the dataset into training and testing datasets using the Train-Test Split technique.
- Applying feature scaling using StandardScaler when required by algorithms such as Logistic Regression and Support Vector Machine.

These preprocessing steps improve model performance by reducing noise, ensuring consistency, and preparing the dataset for accurate prediction.

4. DATA ANALYSIS AND VISUALIZATION

4.1 EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) is a crucial step in any machine learning project. It helps understand the characteristics of the dataset before building predictive models. By performing EDA, hidden patterns, feature relationships, outliers, and data distributions can be identified. These insights improve data quality and help select suitable machine learning algorithms.

The following analyses are performed on the Heart Disease dataset:

- Summary statistics of all numerical features.
- Checking the dimensions of the dataset.
- Identifying missing values and duplicate records.
- Examining data types of each feature.
- Distribution analysis of age, cholesterol, blood pressure, and heart rate.
- Target variable distribution to understand the balance between patients with and without heart disease.
- Correlation analysis among different medical attributes.
- Detection of outliers using box plots.

The dataset is inspected using functions such as `head()`, `info()`, `describe()`, `isnull()`, and `value_counts()` in the Pandas library. These functions provide useful information regarding the dataset structure and quality.

Exploratory analysis shows that most patient records are complete and suitable for machine learning. The statistical summary provides information about the mean, standard deviation, minimum, and maximum values of each feature. Understanding these characteristics helps improve preprocessing and model development.

4.2 DATA VISUALIZATION

Visualization is one of the most effective techniques for understanding complex datasets. Graphical representations make it easier to identify patterns, relationships, and unusual observations.

The following visualizations are used in this project:

HISTOGRAM

Histograms display the frequency distribution of numerical variables such as age, cholesterol level, resting blood pressure, and maximum heart rate. They help identify whether the data is normally distributed or skewed.

The histogram shows that most patients belong to the middle-age group, indicating that heart disease is more common among adults and elderly individuals.

BAR CHART

Bar charts are used to compare the number of patients with and without heart disease. They also help visualize categorical variables such as chest pain type and exercise-induced angina.

The visualization provides a clear understanding of the distribution of the target classes.

BOX PLOT

Box plots are used to identify the spread of numerical variables and detect possible outliers.

The analysis shows that cholesterol and resting blood pressure contain several outliers, which are common in medical datasets. These values are retained because they may represent actual patient conditions rather than errors.

CORRELATION HEATMAP

A correlation heatmap illustrates the relationship between different features.

The analysis indicates:

- Maximum Heart Rate (thalach) has a strong relationship with the target.
- Exercise-Induced Angina (exang) is negatively correlated with heart disease.
- Oldpeak and Chest Pain Type significantly influence prediction.
- Some variables show weak correlation and contribute less to classification.

The heatmap assists in selecting important features during model development.

5. MODEL DEVELOPMENT

Machine Learning algorithms are trained to classify whether a patient is likely to have heart disease based on clinical information.

The dataset is divided into training and testing datasets using an 80:20 ratio. The training dataset is used for learning patterns, while the testing dataset evaluates the model's performance on unseen data.

Several classification algorithms are considered.

5.1 LOGISTIC REGRESSION

Logistic Regression is one of the most widely used algorithms for binary classification problems. Since the target variable has only two possible outcomes (Heart Disease or No Heart Disease), Logistic Regression serves as a simple and effective baseline model.

Advantages:

- Easy to implement.
- Fast training.
- High interpretability.
- Suitable for binary classification.

5.2 DECISION TREE

Decision Tree constructs a tree-like model based on feature values. The algorithm repeatedly splits the dataset into smaller subsets until classification becomes possible.

Advantages:

- Easy to understand.
- Handles both numerical and categorical data.
- Requires little data preprocessing.

However, Decision Trees may overfit the training dataset if not properly controlled.

5.3 RANDOM FOREST

Random Forest is an ensemble learning algorithm that combines multiple Decision Trees to improve prediction accuracy.

Advantages:

- Higher accuracy than a single Decision Tree.
- Reduces overfitting.
- Handles noisy data effectively.
- Provides feature importance scores.

Random Forest is often one of the best-performing algorithms for healthcare datasets.

5.4 SUPPORT VECTOR MACHINE (SVM)

Support Vector Machine is a supervised learning algorithm that identifies the optimal decision boundary separating different classes.

Advantages:

- High accuracy.
- Effective in high-dimensional datasets.
- Good generalization performance.

Feature scaling is usually applied before training SVM models.

7. INTERPRETATION AND DISCUSSION

The primary objective of this project was to develop a machine learning model capable of predicting heart disease using patient medical information. After performing data preprocessing, exploratory data analysis, visualization, model development, and evaluation, several important observations were obtained.

The exploratory data analysis revealed that certain medical attributes have a stronger influence on heart disease prediction than others. Features such as chest pain type, maximum heart rate achieved, exercise-induced angina, ST depression (Oldpeak), number of major vessels (CA), and thalassemia showed significant relationships with the target variable. These attributes play an important role in determining whether a patient is likely to suffer from heart disease.

Visualization techniques such as histograms, box plots, and correlation heatmaps made it easier to understand the distribution of the data and identify relationships among variables. The correlation heatmap indicated both positive and negative correlations between different clinical features. Some variables, such as maximum heart rate and chest pain type, exhibited stronger associations with the target variable than others.

Several machine learning algorithms were considered for classification. Logistic Regression served as an effective baseline model because of its simplicity and interpretability. Decision Tree provided understandable decision rules but was more prone to overfitting. Random Forest improved prediction accuracy by combining multiple decision trees, making it one of the most reliable algorithms for this dataset. Support Vector Machine also performed well after feature scaling and demonstrated good classification capability.

Model evaluation using Accuracy, Precision, Recall, F1-score, Confusion Matrix, and ROC-AUC score provided a comprehensive assessment of model performance. Accuracy alone is insufficient for evaluating medical datasets because false negative predictions can have serious consequences. Therefore, Recall and F1-score are also important metrics when selecting the best predictive model.

Although the machine learning models achieved promising performance, they should not replace professional medical diagnosis. Instead, they should be used as decision-support systems that assist healthcare professionals in identifying high-risk patients and making informed clinical decisions.

Overall, the project demonstrates that machine learning techniques can successfully analyze healthcare data and provide accurate predictions that contribute to early diagnosis and improved patient care.

8. CONCLUSION

Heart disease continues to be one of the leading causes of death worldwide. Early diagnosis and timely treatment can significantly improve patient outcomes. This project demonstrates how machine learning techniques can be applied to healthcare data to predict the likelihood of heart disease using patient clinical information.

The project began with loading and preprocessing the Heart Disease dataset using Python. Exploratory Data Analysis was performed to understand the dataset structure, identify missing values, analyze statistical properties, and visualize feature distributions. These analyses provided valuable insights into the relationships among different medical variables.

Several supervised machine learning algorithms were studied and evaluated. The models were assessed using widely accepted evaluation metrics such as Accuracy, Precision, Recall, F1-score, Confusion Matrix, and ROC-AUC score. These evaluation techniques ensured that the predictive performance of the models was measured comprehensively.

The results indicate that machine learning can effectively classify patients based on their medical characteristics and assist healthcare professionals in identifying individuals at higher risk of heart disease. Such systems have the potential to reduce diagnosis time, improve decision-making, and support preventive healthcare.

In conclusion, this project highlights the importance of Artificial Intelligence and Machine Learning in modern healthcare. Although predictive models cannot replace experienced doctors, they provide valuable support in clinical decision-making and can contribute to better patient management.

9. FUTURE SCOPE

The Heart Disease Prediction System can be further improved by incorporating advanced techniques and larger datasets. Future enhancements may include:

- Collecting larger and more diverse patient datasets to improve model generalization.
- Applying Deep Learning algorithms such as Artificial Neural Networks (ANNs) and Convolutional Neural Networks (CNNs) for improved predictive performance.
- Performing Hyperparameter Optimization using Grid Search and Random Search to improve model accuracy.
- Implementing Cross-Validation techniques for more reliable model evaluation.
- Developing ensemble models by combining multiple classifiers to increase robustness.
- Integrating Explainable Artificial Intelligence (XAI) techniques to improve transparency and help healthcare professionals understand model predictions.
- Deploying the trained model as a web application using Flask or Django for real-time prediction.
- Developing a mobile healthcare application that enables users to assess heart disease risk remotely.

- Integrating electronic health records (EHR) to provide automatic prediction using hospital databases.
- Extending the system to predict other cardiovascular diseases and chronic health conditions.

These improvements can make the prediction system more accurate, reliable, user-friendly, and suitable for practical healthcare applications.

REFERENCES

1. Python Software Foundation. **Python Documentation**. <https://docs.python.org/>
2. Pandas Development Team. **Pandas Documentation**. <https://pandas.pydata.org/docs/>
3. NumPy Developers. **NumPy Documentation**. <https://numpy.org/doc/>
4. Matplotlib Development Team. **Matplotlib Documentation**. <https://matplotlib.org/stable/>
5. Seaborn Documentation. <https://seaborn.pydata.org/>
6. Scikit-learn Developers. **Scikit-learn User Guide**. <https://scikit-learn.org/stable/>
7. UCI Machine Learning Repository. **Heart Disease Dataset**.
8. Aurélien Géron. **HANDS-ON MACHINE LEARNING WITH SCIKIT-LEARN, KERAS & TENSORFLOW**. O'Reilly Media.
9. Ian Goodfellow, Yoshua Bengio, Aaron Courville. **DEEP LEARNING**. MIT Press.
10. World Health Organization (WHO). **Cardiovascular Diseases Fact Sheets**.

6. MODEL EVALUATION

After training, the machine learning models are evaluated using standard classification metrics.

ACCURACY

Accuracy measures the percentage of correctly classified patient records.

$$\text{Accuracy} = (\text{Correct Predictions} / \text{Total Predictions}) \times 100$$

Higher accuracy indicates better predictive performance.

PRECISION

Precision measures how many predicted positive cases are actually positive.

High precision reduces false positive predictions and improves reliability.

RECALL

Recall measures how many actual heart disease patients are correctly identified by the model.

High recall is particularly important in medical diagnosis because missing a diseased patient can have serious consequences.

F1-SCORE

F1-Score combines Precision and Recall into a single metric.

It provides balanced evaluation, especially when class distributions are uneven.

CONFUSION MATRIX

The Confusion Matrix summarizes prediction results using four categories:

- True Positive (TP)

- True Negative (TN)
- False Positive (FP)
- False Negative (FN)

It provides detailed insight into classification performance and helps identify prediction errors.

ROC CURVE AND AUC SCORE

The Receiver Operating Characteristic (ROC) Curve illustrates the trade-off between sensitivity and specificity.

The Area Under the Curve (AUC) measures overall classification performance.

- AUC close to **1.0** indicates excellent prediction.
- AUC around **0.5** indicates poor performance similar to random guessing.

The evaluation metrics help compare multiple machine learning algorithms and identify the most suitable model for heart disease prediction.